Information Systems and Databases 2025/2026

Project Assignment - Part 1

In this Part 1 of the project assignment, you will design a database model to answer the information requirements of an application whose domain is presented below. Your job is to deliver a concise and clean data model using the Entity-Association model graphic notation taught in class and, along with it, identify and specify the appropriate Integrity Constraints.

In addition you will create, populate, and query a database that implements the Entity-Association model presented.

1. Domain Description

You have been asked to design an information system to support a Boating Management System with the following information requirements:

The system stores essential information about boats. Each boat has a name (not necessarily unique), a length, and a unique national boat identifier (the 'cni') assigned by the country where it is registered. The year of registration must also be recorded. Every boat belongs to a class (e.g., "Class 1", "Class 2"), and each class has an associated maximum allowed length.

Each sailor has a first name, surname, and email, and the system must allow any of these attributes to be updated. Every sailor is categorized as either Senior or Junior. Senior sailors have additional responsibilities (described below), which the system must enforce.

Sailors often make joint reservations to use a boat for a specific time interval, defined by a start date and an end date. All participating sailors become the authorized sailors for that reservation, and every reservation must include at least one authorized sailor.

Only authorized sailors may navigate or operate the boat during the reservation's time frame. Among the authorized sailors, one must be a Senior sailor who serves as the responsible person for the reservation.

A boat may support multiple trips under a single reservation.

Some users want the reservation to be fast and to see their boat picture on the screen.

Each trip associated with a reservation has a skipper (who must be one of the reservation's authorized sailors), a start location, an end location, a take-off date, an arrival date, and an *insurance reference. Every location in the system includes a name, latitude, and longitude. In the system any two locations must be at least one nautical mile apart.*

Each trip must also explicitly record the list of water jurisdictions that will be navigated, since boats must display the flags of all such countries. Jurisdictions are zones over which countries have legal authority or 'International Watters' which do not have any administering country.

When sailing from one location to another, the boat will cross multiple jurisdictions even within the same country.

Example: *Sailing from Faro (Portugal) to Cádiz (Spain)*

1. *Portuguese Internal Waters*
2. *Portuguese Territorial Sea*
3. *Portuguese EEZ*
4. *International Waters*
5. *Spanish EEZ*
6. *Spanish Territorial Sea*
7. *Spanish Internal Waters*

A country has a unique name, a unique flag, and a unique standard ISO code. Any country that registers boats must have at least one location defined.

Sailors may hold personal sailing certifications. Each certification has an issue date and an expiry date, and authorizes the sailor to act as skipper for a specific boat class in one or more country jurisdictions.

2. Work to be developed

2.1 Entity-Association Model

1. Design an **Entity-Association model diagram** for the problem domain presented in the previous section.

2. Identify those situations that are inconsistent in the problem domain, but that are allowed in the presented Entity-Association model, and **define a set of Integrity Constraints** that complete the proposed model in order to prohibit situations that are invalid.

2.2 Translation to SQL

1. Using the SQL (DDL) language, present the statements to create the database **corresponding to the Entity-Association model developed above**. Ensure that data types and field sizes selected are the most appropriate.

2. The constraints on each field, row and table must also be specified. Ensure that **NOT NULL**, **CHECK**, **PRIMARY KEY**, **UNIQUE**, and **FOREIGN KEY** constraints are appropriately used. The use of accented characters and cedillas should be avoided.

3. List all the constraints that exist in the Entity-Association model that cannot be captured (implemented) in the SQL schema, writing them **as comments** to the respective tables.

2.3 Database Loading

Define in SQL (DML) all the **INSERT** statements that you find necessary to cover specific characteristics that real data scenarios can have in order to validate the expected results of the queries below (keep in mind that each table should contain at least one row and that every query will have to return a non-empty result).

2.4 Simple SQL Queries

A. The name of all boats that <u>are used</u> in some trip.
B. The name of all boats that <u>are not used</u> in any trip.
C. The name of all boats registered in 'PRT' (ISO code) for which at least one responsible for a reservation has a surname that ends with 'Santos'.
D. The full name of all skippers without any certificate corresponding to the class of the trip's boat.

3. Grading

The project will be graded from the submission that should contain all the answers to the items requested above. Oral discussions may be requested to randomly selected groups. The following table indicates the valuation of each part of the work to be developed.

Item	**Grading (0-20)**
Entity-Association Model	8.0
Integrity Constraints of the E-A Model	2.0
Translation to SQL	3.0
Integrity Constraints in SQL	2.0

Database Loading	2.0
SQL Queries	3.0

IMPORTANT - READ CAREFULLY

- The Entity-Association model must be **expressed** in **the notation taught in class**
- The diagram should be drawn electronically on a diagraming tool. Refer to the lecture slides suggestions with suggestions of diagramming tools that can be used.
- The Integrity Constraints to the Entity-Association model must be written as **assertions expressed in terms of the concepts in the Entity-Association model**, that is, in terms of attributes, entities, and relationships between them.
- The "cognitive-efficiency" as perceived in terms of **conciseness** and **cleanliness** of the Entity-Association model will be **evaluated.**
- The **quality** (in terms of **organisation**, **indentation**, and **simplicity**) of the **SQL scripts** will also be evaluated.
- The presence of SQL instructions **<u>that cannot be accurately explained</u>** by the students will negatively impact the grade.

⚠⚠ ***The teachers have tested ChatGPT and found it <u>to be incapable</u> of analyzing the problem domain and producing an acceptable E-A model.*** ⚠ ⚠

4. Report format and submission

The submission must be a structured **zip** file named **`submission-sidb-01-GG.zip`**[1] where GG is the number of the group as follows[2]:

`cover.pdf`	**A cover page PDF with the names of the authors and any notes** The report should start with a cover page with the title "**SIBD Project - Part 1**", with the **name and number of students**, the **relative percent of each student's contribution**, together with the **approximate total effort (in hours)**

[1] ⚠ Only ZIP or GZ formats are accepted. Other archive formats (such as RAR) are not accepted.
[2] Penalties will apply to the submissions that do not meet the structure requested.

	that each element of the group dedicated to the project, the **number of the group**, and **the name of the laboratory teacher.** Length: The report shall have a **maximum** of **2 pages** including the cover page.
`schema.sql`	**File with the schema creation instructions** This file should cleanly drop any existing tables before recreating them. The constraints that exist in the Entity-Association model that are not captured (implementable) in the SQL schema should be added as comments to the corresponding tables. The script should be runnable in POSTGRES on `db.tecnico.ulisboa.pt`.
`queries.sql`	**File with the SQL Queries** Please ensure that queries are clearly marked using comments. The script should be runnable in POSTGRES on `db.tecnico.ulisboa.pt`.
`populate.sql`	**File with the scripts that populate the database** The script should be runnable in POSTGRES on `db.tecnico.ulisboa.pt`.
`output.txt`	**File with the output of each query** Please make sure that the output of each query is not empty.

NOTE: Penalties will apply to groups that do not follow the submission instructions. Evaluation elements that are not found as prescribed as above **will not be taken into account for grading purposes**. DO NOT wait until the last moment to submit. Late submissions will NOT be accepted.